

QTM 220

Regression with Randomized Experiments

November 29, 2017

Outline

Simple Regression

Multiple Regression

Interactive Regression

Logistic Regression

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Multiple Regression

Interactive Regression

Logistic Regression

Linear regression of Y on T

With binary T , the slope is a difference in means,

$$E[Y|T = t] = \beta_0 + \beta_1 \cdot t$$

$$E[Y|T = 0] = \beta_0$$

$$E[Y|T = 1] = \beta_0 + \beta_1$$

and if the potential outcome are well defined such that $Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0)$ and T has been randomized,

$$\begin{aligned}\beta_1 &= E[Y_i|T_i = 1] - E[Y_i|T_i = 0] \\ &= E[Y_i(1)|T_i = 1] - E[Y_i(0)|T_i = 0] \\ &= E[Y_i(1)] - E[Y_i(0)] \\ &= E[Y_i(1) - Y_i(0)]\end{aligned}$$

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What is the population?

When the sample is a sample from a population

- the units of analysis are approximately an i.i.d. sample from a population (e.g., some surveys)

When the sample is the population

- sample is all states in the country (e.g., effect of voter id laws on Trump vote share)
- sample is all voters on the voters' register (e.g., effect of GOTV on turnout)
- sample is provisionally taken to be the population in order to focus first on internal validity (and postpone consideration of external validity)

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Point Estimation with Linear Regression

If you have an i.i.d. sample, a binary treatment is randomized, and you run a linear regression of Y on T , you get

- $\hat{\beta}_1$ is an unbiased and consistent estimator of the ATE
- because this is the ATE for the population (from which the i.i.d. sample comes), we sometime refer to it as the PATE (population average treatment effect)

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Inference with an i.i.d. sample

If you have an i.i.d. sample, a binary treatment is randomized, and you use robust standard errors or bootstrapping, then the CIs, tests, and p -values we developed for descriptive inference with a simple linear regression will be approximately correct.

If homoskedasticity (constant conditional variance) also holds, then the CIs, tests, and p -values derived the $lm()$ function (or similar) will also will be approximately correct.

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Inference when sample is the population

Because sample is the population, uncertainty due solely to random assignment of treatment. Fortunately, the CIs, tests, and p -values we developed for descriptive inference will be conservative in the sense that

- $(1 - \alpha)\%$ CIs will have greater than $(1 - \alpha)\%$ coverage.
- α level tests will actually have probabilities of Type I error less than α .
- Reported p -values will be larger than the true p -values.
- ... still need to worry about possible heteroskedasticity.

Potential Outcomes and the Classical Econometric Linear Model

$$\begin{aligned}Y_i &= T_i Y_i(1) + (1 - T_i) Y_i(0) \\&= Y_i(0) + (Y_i(1) - Y_i(0)) T_i \\&= \beta_{0i} + \beta_{1i} T_i \\&= \beta_0 + \beta_{1i} T_i + \epsilon_i\end{aligned}$$

If effects are constant across individuals (i.e., $\beta_{1i} = \beta_1$ for all i), then

$$\begin{aligned}Y_i &= \beta_0 + \beta_{1i} T_i + \epsilon_i \\&= \beta_0 + \beta_1 T_i + \epsilon_i\end{aligned}$$

This is the linear model from ALZ page 125.

Unconfoundedness vs Errors Terms

With the previous definitions and constant effects, we can re-write the potential outcomes as the following:

$$Y_i(1) = \beta_0 + \beta_1 + \epsilon_i$$

$$Y_i(0) = \beta_0 + \epsilon_i$$

This allows us to relate the unconfoundedness assumption to the classical error terms assumption.

$$E[Y_i(1)] = E[Y_i(1) | T = 1]$$

$$\beta_0 + \beta_1 + E[\epsilon_i] = \beta_0 + \beta_1 + E[\epsilon_i | T = 1]$$

Therefore we have unconfoundedness when Assumption 5 from ALZ page 128 holds (i.e., $Cov(T_i, \epsilon_i) = 0$ for all i)

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$$E[Y_i(0)] = E[Y_i(0) | T = 1]$$

$$\beta_0 + E[\epsilon_i] = \beta_0 + E[\epsilon_i | T = 1]$$

Therefore we have unconfoundedness when Assumption 5 from ALZ page 128 holds (i.e., $Cov(T_i, \epsilon_i) = 0$ for all i)

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Multiple Regression

Interactive Regression

Logistic Regression

Recall: Randomization Implies Balance

Suppose the subjects were randomly assigned to $T = 0$ and $T = 1$ (also imagine these six subjects actually represent 600 subjects), and we measure a “pre-treatment” covariate (i.e., not affected by T).

i	X_i	T_i	Y_i	$Y_i(1)$	$Y_i(0)$	$Y_i(1) - Y_i(0)$
1	0	1	70	70	?	?
2	1	1	80	80	?	?
3	1	0	60	?	60	?
4	0	0	70	?	70	?
5	0	0	70	?	70	?
6	1	0	80	?	80	?

Randomization implies that X is independent of T , which implies that $E[X|T = 1] = E[X|T = 0]$.

Balance on Pre-treatment Variables in the Gerber, Green, and Larimer Experiment

TABLE 1. Relationship between Treatment Group Assignment and Covariates (Household-Level Data)

	Control	Civic Duty	Hawthorne	Self	Neighbors
	Mean	Mean	Mean	Mean	Mean
Household size	1.91	1.91	1.91	1.91	1.91
Nov 2002	.83	.84	.84	.84	.84
Nov 2000	.87	.87	.87	.86	.87
Aug 2004	.42	.42	.42	.42	.42
Aug 2002	.41	.41	.41	.41	.41
Aug 2000	.26	.27	.26	.26	.26
Female	.50	.50	.50	.50	.50
Age (in years)	51.98	51.85	51.87	51.91	52.01
N =	99,999	20,001	20,002	20,000	20,000

Note: Only registered voters who voted in November 2004 were selected for our sample. Although not included in the table, there were no significant differences between treatment group assignment and covariates measuring race and ethnicity.

Why balance in large samples?

i	X_i	T_i	Y_i	$Y_i(1)$	$Y_i(0)$	$Y_i(1) - Y_i(0)$
1	0	1	70	70	?	?
2	1	1	80	80	?	?
3	1	0	60	?	60	?
4	0	0	70	?	70	?
5	0	0	70	?	70	?
6	1	0	80	?	80	?

- The $T = 1$ rows are a random sample of the rows, therefore $E[X|T = 1] = E[X]$.
- The $T = 0$ rows are a random sample of the rows, therefore $E[X|T = 0] = E[X]$.
- Therefore $E[X|T = 1] = E[X|T = 0]$.

Warning: Balance only for pre-treatment X

- If X is affected by T , then the previous results will not hold
- Even though the $T = 1$ rows are a random sample of the rows, $E[X|T = 1] \neq E[X]$.
- Even though the $T = 0$ rows are a random sample of the rows, $E[X|T = 0] \neq E[X]$.
- Therefore $E[X|T = 1] \neq E[X|T = 0]$.
- Suppose Y indicates a heart attack, T is a blood pressure medicine, and X is blood pressure.

Recall: What if x and z are uncorrelated?

```
> z <- rnorm(n)
> x <- rnorm(n) # no z
> cor(x,z)
[1] 0.01075281
> y <- x + z + rnorm(n)
> lm(y ~ x)
Coefficients:
(Intercept)                x
      0.01027          1.00940
```

```
> lm(y ~ x + z)
```

```
Coefficients:
(Intercept)                x                z
      0.00417          0.99901          0.98509
```

Including Covariates in the Gerber, Green, and Larimer Experiment

TABLE 3. OLS Regression Estimates of the Effects of Four Mail Treatments on Voter Turnout in the August 2006 Primary Election

	Model Specifications		
	(a)	(b)	(c)
Civic Duty Treatment (Robust cluster standard errors)	.018* (.003)	.018* (.003)	.018* (.003)
Hawthorne Treatment (Robust cluster standard errors)	.026* (.003)	.026* (.003)	.025* (.003)
Self-Treatment (Robust cluster standard errors)	.049* (.003)	.049* (.003)	.048* (.003)
Neighbors Treatment (Robust cluster standard errors)	.081* (.003)	.082* (.003)	.081* (.003)
N of individuals	344,084	344,084	344,084
Covariates**	No	No	Yes
Block-level fixed effects	No	Yes	Yes

Note: Blocks refer to clusters of neighboring voters within which random assignment occurred. Robust cluster standard errors account for the clustering of individuals within household, which was the unit of random assignment.

* $p < .001$.

** Covariates are dummy variables for voting in general elections in November 2002 and 2000, primary elections in August 2004, 2002, and 2000.

Simple vs Additive Multiple Linear Regression

Analysis of Randomized Experiments

- In very large samples, additive multiple linear regression will produce approximately the same answer as simple linear regression for the ATE (e.g., social pressure).
- In medium/large samples, (> 30 d.f.), multiple linear regression will have small bias but will generally more efficient than simple linear regression.
- In small samples, the bias in multiple linear regression will be too much.

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Interpreting the Multiple Regression Model

Consider the multiple regression model from ALZ page 158.

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + \epsilon_i$$

- Should we interpret both β_1 and β_2 as causal effects?
- What randomized experiment would justify this interpretation?

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With a randomized experiment, the unconfoundedness assumption will hold conditional on any set of pre-treatment covariates (X).

$$\begin{aligned} E[Y_i(1)|X = x] &= E[Y_i(1)|T_i = 1, X_i = x] \\ &= E[Y_i|T_i = 1, X_i = x] \end{aligned}$$

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Conditional Average Treatment Effect (CATE)

$$E[Y_i(1) - Y_i(0)|X_i = x]$$

CATE examples

- average effect among men/average effect among women
- average effect among those with annual income of 15k,16k,17k,18k,....
- average effect among low/high propensity voters

Estimating CATEs with Interactions

With a binary T and a binary X , we can always write the conditional expectation as a linear interactive function,

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Interpretation with Interactive Model

Therefore, with an interactive model,

$$\begin{aligned}E[Y|T = 1, X = 0] - E[Y|T = 0, X = 0] &= (\beta_0 + \beta_1) - \beta_0 \\ &= \beta_1\end{aligned}$$

can be interpreted as the CATE among the $X = 0$ subjects.

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$$\begin{aligned}E[\beta_1 + \beta_3 \cdot X] &= \beta_1 + \beta_3 \cdot E[X] \\ &= (1 - E[X])\beta_1 + E[X](\beta_1 + \beta_3)\end{aligned}$$

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Interpretation with Interactive Model (non-binary X)

With non-binary X , a linear interactive model requires a linearity assumption

$$\begin{aligned}E[Y|T = t, X = x] &= \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x + \beta_3 \cdot t \cdot x \\ &= \beta_0 + \beta_2 \cdot x + (\beta_1 + \beta_3 \cdot x) \cdot t\end{aligned}$$

where $(\beta_1 + \beta_3 \cdot x)$ can be interpreted as the CATE among the $X = x$ subjects.

And the ATE can still be recovered in the following way

$$E[\beta_1 + \beta_3 \cdot X] = \beta_1 + \beta_3 \cdot E[X]$$

... more complicated for non-linear models.

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CATEs with Additive Linear Regression

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Conditional Average Treatment Effect (CATE)

$$E[Y_i|T_i = 1, X_i = x] - E[Y_i|T_i = 0, X_i = x] = \beta_1$$

CATEs with Interactive Linear Regression

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CATEs with Interactive Linear Regression

$$\begin{aligned}E[Y_i(1)|X = x] &= E[Y_i(1)|T_i = 1, X_i = x] \\&= E[Y_i|T_i = 1, X_i = x] \\&= \beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot x + \beta_3 \cdot 1 \cdot x\end{aligned}$$

$$\begin{aligned}E[Y_i(0)|X_i = x] &= E[Y_i(0)|T_i = 0, X_i = x] \\&= E[Y_i|T_i = 0, X_i = x] \\&= \beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot x + \beta_3 \cdot 0 \cdot x\end{aligned}$$

Conditional Average Treatment Effect (CATE)

$$E[Y_i|T_i = 1, X_i = x] - E[Y_i|T_i = 0, X_i = x] = \beta_1 + \beta_3 x$$

CATEs with Logistic Regression

$$\begin{aligned} E[Y_i(1)|X = x] &= E[Y_i(1)|T_i = 1, X_i = x] \\ &= E[Y_i|T_i = 1, X_i = x] \\ &= \frac{\exp(\beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot x)}{1 + \exp(\beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot x)} \end{aligned}$$

CATEs with Logistic Regression

$$\begin{aligned} E[Y_i(1)|X = x] &= E[Y_i(1)|T_i = 1, X_i = x] \\ &= E[Y_i|T_i = 1, X_i = x] \\ &= \frac{\exp(\beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot x)}{1 + \exp(\beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot x)} \end{aligned}$$

$$\begin{aligned} E[Y_i(0)|X_i = x] &= E[Y_i(0)|T_i = 0, X_i = x] \\ &= E[Y_i|T_i = 0, X_i = x] \\ &= \frac{\exp(\beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot x)}{1 + \exp(\beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot x)} \end{aligned}$$

CATEs with Logistic Regression

$$\begin{aligned}E[Y_i(1)|X = x] &= E[Y_i(1)|T_i = 1, X_i = x] \\&= E[Y_i|T_i = 1, X_i = x] \\&= \frac{\exp(\beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot x)}{1 + \exp(\beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot x)}\end{aligned}$$

$$\begin{aligned}E[Y_i(0)|X_i = x] &= E[Y_i(0)|T_i = 0, X_i = x] \\&= E[Y_i|T_i = 0, X_i = x] \\&= \frac{\exp(\beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot x)}{1 + \exp(\beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot x)}\end{aligned}$$

Conditional Average Treatment Effect (CATE)

$$\begin{aligned}E[Y_i|T_i = 1, X_i = x] - E[Y_i|T_i = 0, X_i = x] &= \\ \frac{\exp(\beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot x)}{1 + \exp(\beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot x)} - \frac{\exp(\beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot x)}{1 + \exp(\beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot x)}\end{aligned}$$

ATEs with Logistic Regression

An ATE can be recovered in the following way

$$E_X \left[\frac{\exp(\beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot x)}{1 + \exp(\beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot x)} - \frac{\exp(\beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot x)}{1 + \exp(\beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot x)} \right]$$

Estimating ATEs with Logistic Regression

$$\widehat{ATE} = \frac{1}{n} \sum_i^n \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot x_i)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot x_i)} - \frac{1}{n} \sum_i^n \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot x_i)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot x_i)}$$

How would you do this in *R*? Standard errors?

CATEs and ATEs with other

$$\begin{aligned} E[Y_i(1)|X = x] &= E[Y_i(1)|T_i = 1, X_i = x] \\ &= E[Y_i|T_i = 1, X_i = x] \\ &=? \end{aligned}$$

CATEs and ATEs with other

$$\begin{aligned} E[Y_i(1)|X = x] &= E[Y_i(1)|T_i = 1, X_i = x] \\ &= E[Y_i|T_i = 1, X_i = x] \\ &=? \end{aligned}$$

$$\begin{aligned} E[Y_i(0)|X_i = x] &= E[Y_i(0)|T_i = 0, X_i = x] \\ &= E[Y_i|T_i = 0, X_i = x] \\ &=? \end{aligned}$$

CATEs and ATEs with other

$$\begin{aligned} E[Y_i(1)|X = x] &= E[Y_i(1)|T_i = 1, X_i = x] \\ &= E[Y_i|T_i = 1, X_i = x] \\ &=? \end{aligned}$$

$$\begin{aligned} E[Y_i(0)|X_i = x] &= E[Y_i(0)|T_i = 0, X_i = x] \\ &= E[Y_i|T_i = 0, X_i = x] \\ &=? \end{aligned}$$

$$E_X[?]$$